



BASICS OF CIVIL ENGINEERING MODULE I



SYLLABUS

- General introduction to civil engineering: relevance of civil engineering in the overall infrastructural development of the country. Responsibility of an engineer in ensuring the safety of built environment.
- Brief introduction to major disciplines of civil engineering like transportation engineering, structural engineering, geo-technical engineering, water resources engineering and environmental engineering.
- Introduction to buildings: types of buildings, selection of site for buildings, components of a residential building and their functions.
- Building rules and regulations: relevance of nbc, kbr & crz norms (brief discussion only). Building area: plinth area, built up area, floor area, carpet area and floor area ratio for a building as per KBR.

CIVIL ENGINEERING

- Oldest Engineering discipline
- Civil Engineers plan, design, construct and operate the facilities essential to modern life –buildings, bridges, highways, water treatment plant etc.
- Civil Engineers are problem solvers – meet challenges of pollution, traffic congestion, drinking water and energy needs, urban redevelopment, and community planning.

HISTORY

- Oldest Engineering discipline



HISTORY

- Word “Military Engineer” originated first-undertook works related to Military operations during war
- Engineer undertaking works other than military engineering-“Civil Engineer”
- History of Civil Engineering is a mirror of the history of human beings on earth

Relevance of Civil Engineering

- One of the world’s most important jobs
- Build and improve our quality of life
- Plan, design, construct and operate facilities essential to modern life



Relevance of Civil Engineering

- All construction work of large scale needs guidance of a civil engineer for the project planning
- Project planning consists of
 - Conception
 - Feasibility of work
 - Design
 - Construction
 - Maintenance



Relevance of Civil Engineering in Infrastructural Development of the country(UQ)

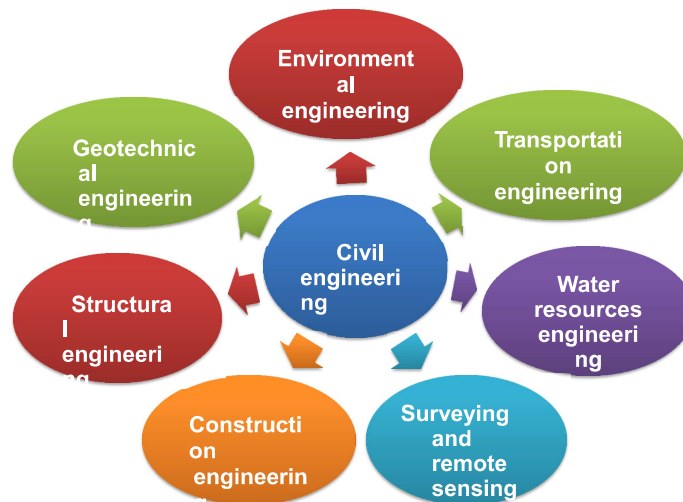
- Infrastructure – framework of supporting system
- Consists of transportation, communication, energy, lifeline facilities, irrigation facilities etc
- Helps a community to function, grow and develop

- Infrastructure development-contributes-economic development of nation
- Higher infrastructure-higher will be growth
- Infrastructure deals with:
 - Transportation networks
 - Communication networks
 - Energy sector
 - Education facility
 - Good health care facility etc

Role of Civil Engineer in Infrastructural Development of the country(UQ)

- Construction of roads, railway, ports, harbors and airports
- Construction of dams and proper utilization of water resources.
- Construction of Housing, commercial and industrial complexes, Maintenance of facility
- Rebuilding, Rehabilitation, Retrofitting and Repair

DISCIPLINES OF CIVIL ENGINEERING



VARIOUS DISCIPLINES

- Construction Engineering
- Structural Engineering
- Geotechnical Engineering
- Environmental Engineering
- Transportation Engineering
- Water resources Engineering
- Surveying and Remote Sensing

CONSTRUCTION ENGG

- Turns designs into reality
- Management of construction resources: labour, materials, equipment, money and time.
- Application of knowledge of construction methods and equipment, along with the principles of financing, planning and management, to turn the designs into successful facilities.



STRUCTURAL ENGG

- Analyze and design of structures that support their own weight and the loads they carry and resist extreme forces from wind, earthquakes, bombings, temperature & others.
- Structural Engineers develop appropriate combinations of steel, concrete, timber, plastic & new exotic materials
- Must also consider the economics,
- aesthetics and social implications of their creations.



GEOTECHNICAL ENGG

- Develop projects below the ground. Eg: tunnels, foundation, offshore platforms, and containment structures for solid and liquid wastes.
- Analyze the properties of soil and rock that support and affect the behavior of these structures.
- Evaluate potential settlements of buildings, the stability of slopes and fills, seepage of ground water and effects of earthquakes.

GEOTECHNICAL ENGG



- Determine the best way to support a structure in the ground.
- Take part in design and construction of dams, embankments and retaining walls.



ENVIRONMENTAL ENGG



- Resolve the problems of providing safe drinking water, cleaning up contaminated sites with hazardous materials, disposing of waste water and managing solid wastes.
- Translate physical, chemical & biological processes into systems to destroy toxic substances, remove pollutants from water, reduce non- hazardous solid wastes, etc.



TRANSPORTATION ENGG

- Plan, design, construct, maintain and operate various transportation facilities(e.g., highway, railway, airports, ports, etc.).
- Upgrading our transportation capability by improving traffic control and mass transit systems.



WATER RESOURCES ENGG

- Deals with physical control of water
- Work with others to prevent floods, supply of water, protect beaches or to manage redirect rivers
- Design, construct and maintain hydroelectric power facilities, dams, pipelines, pumping stations, etc.



SURVEYING & REMOTE SENSING

- Surveying is to determine the positions of points on, above below surface of the earth by means of direct or indirect measurements of distances, elevations and directions.
- Surveying has advanced from chain surveying to remote sensing with the advent of various electronic sophisticated instruments.



INTRODUCTION TO TYPES OF BUILDING AS PER NBC

NBC

- National Building Code is prepared to unify the building regulations throughout the country.
- NBC is prepared by Bureau of Indian Standards (BIS).
- First version is published in 1970
- This is revised in 1983. Thereafter three major amendments were issued to the 1983 version, two in 1987 and the third in 1997
- Later this edition was again revised to current one in 2005, to which two amendments were issued in 2015.
- Third revision in 2016

TYPES OF BUILDING

- Classification based on occupancy
- Classification based on fire resistance
- Classification based on height
- Classification based on load transfer
- Classification based on material used

TYPES OF BUILDING

Classification based on occupancy

- Group A: Residential
- Group B: Educational
- Group C: Institutional
- Group D: Assembly
- Group E: Business
- Group F: Mercantile
- Group G: Industrial
- Group H: Storage
- Group J: Hazardous

Group A: Residential Buildings

Buildings in which sleeping accommodation provided for normal residential purposes with or without cooking or dining or both facilities, except any building classified under Group C.



Group B: Educational Buildings

➤ Buildings used for school, college, other training institutions, for day- care purposes involving assembly for instruction, education or recreation for not less than 20 students.

➤ Residential accommodation provided in school/ institution are classified as sub-division A-3.



Group C: Institutional Buildings

➤ Any building or part thereof, which is used for purposes, such as **medical or other treatment or care of persons** suffering from physical or mental illness, disease or infirmity; cares of infants, convalescents or aged persons and for penal or correctional detention in which the liberty of inmates is restricted.



Group D: Assembly Buildings

- Any building or part of building, where number of persons **not less than 50 congregate or gather** for amusement, recreation, social, religious, patriotic, civil, travel and similar purposes.
- Eg: theatres, assembly hall, museum, gymnasiums, restaurants, places for worships etc



Group E: Business Buildings

- Any building or part of a building, used for **transaction of business** (other than that covered by building in group F), **for keeping accounts and records** and similar purposes.
- Barber shops, beauty parlours are included in this group.



Group F: Mercantile Buildings

- Any building or part of a building, used as shops, stores, market, for display and sale of merchandise, either wholesale or retail



Group G: Industrial Buildings

- Any building or part of a building or structure, in which products or materials of all kinds and properties are fabricated, assembled, manufactured or processed.
- Assembly plants, industrial laboratories, dry cleaning plants, power plants generating units, pumping stations, laundries, building or structures in gas plants, etc.



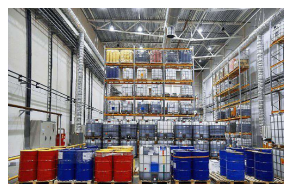
Group H: Storage Buildings

- Any building or a part of a building used primarily for or the **storage or sheltering** (including servicing, processing, or repairing incidental to storage) **of goods, wares or merchandise** (except that those involve highly combustible or explosive products or materials) **vehicles or animals**.
- Warehouses, cold storage, grain elevators, freight depots, barns, transit sheds, storehouse, garage, hangers etc



Group J: Hazardous Buildings

- Used for the storage, handling, manufacture or processing of highly combustible explosive materials, which are liable to burn with extreme rapidity and which may produce poisonous fumes or explosions for storage, handling, manufacturing or processing which involve **highly corrosive, toxic alkalis, acids or other liquids or chemical producing flames, fumes and explosive, poisonous, irritant or corrosive gases**.



Occupancy classification of buildings

Group	Occupancy	Sub divisions
A	Residential	A1 Lodging or rooming houses A2 One or two family dwellings A3 Dormitories A4 Apartments A5 Hotels
B	Educational	Building used for colleges, schools, day care purpose involving assembly for instructions, education etc

Occupancy classification of buildings

Group	Occupancy	Sub divisions
C	Institutional	C1 Hospitals and sanatoria C2 Homes for aged, orphans C3 Prisons, mental hospitals
D	Assembly	D1 Theatres with fixed seats for more than 1000 persons D2 Theatres with fixed seats for less than 1000 persons D3 Halls with capacity to hold more than 300 persons but with no permanent seating

Occupancy classification of buildings

Group	Occupancy	Sub divisions
D	Assembly	D4 Halls for less than 300 persons but with no permanent seating D5 All other assembly buildings not covered above
E	Business	E1 Office, banks, professional establishments for doctors, engineers, lawyers E2 Laboratories and research E3 Computer installations

Occupancy classification of buildings

Group	Occupancy	Sub divisions
F	Mercantile	F1 Shops, stores, markets with area up to 500m ² with storage incidental to sales F2 Underground shopping centres/ departmental stores with area more than 500m ² with storage and services incidental to sales
G	Industrial	G1 Buildings used for low hazard industries G2 Buildings used for moderate hazard industries G3 Buildings used for high hazard industries

Occupancy classification of buildings

Group	Occupancy	Sub divisions
H	Storage buildings	Used for storage or sheltering of goods, vehicles, animals Warehouses, grain stores, garages, stables
Ij	Hazardous	Used for storage, handling, manufacturing and processing of corrosive, toxic, inflammable, explosive materials

CLASSIFICATION BASED ON FIRE RESISTANCE

- Type 1: all structural components - incombustible ,fire resistant for 4 hrs
- Type 2: all structural components -incombustible ,fire resistant for 3 hrs
- Type 3: exterior portion – incombustible components, inner parts other than inner walls combustible materials. Fire resistant for 2 hrs
- Type 4: made of wood or other combustible materials. Fire resistant for 1 hr

- **CLASSIFICATION BASED ON HEIGHT**

- High rise: height >15 m
- Low rise: height <15 m

- **CLASSIFICATION BASED ON LOAD TRANSFER**

- Load bearing masonry building
- Framed building

- **CLASSIFICATION BASED ON MATERIAL USED**

- Earthen
- Masonry
- Wooden
- Steel
- RCC

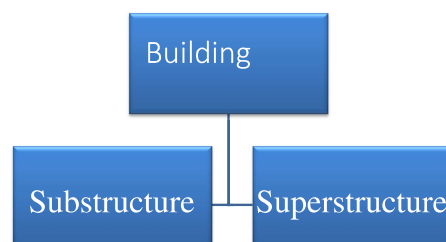


SELECTION OF SITE FOR BUILDINGS

- Level ground with good quality of soil
- Calm but reasonably developed location
- Road connectivity- and other modes of transport
- Good communication facilities
- Electricity, water and sewer lines should be available
- Away from hazardous industries
- Flood prone areas, water logged areas and reclaimed land should be avoided
- Good natural ventilation and lighting

- The amenities like schools, recreation centers, shopping centers and hospitals should be there
- Drainage properties
- Regular shape with sufficient frontage
- The area must be sufficient for present and future development
- The places prone to air and water pollution should be avoided
- Proper ownership and other legal matters have to be checked before buying a site
- The type of land use recommended at proposed site should be checked with local authorities

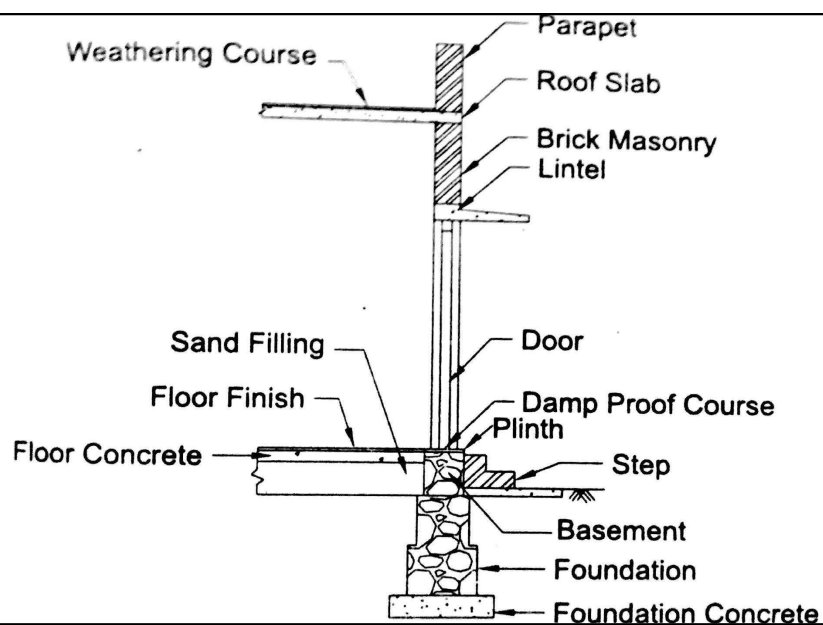
COMPONENTS OF A RESIDENTIAL BUILDING

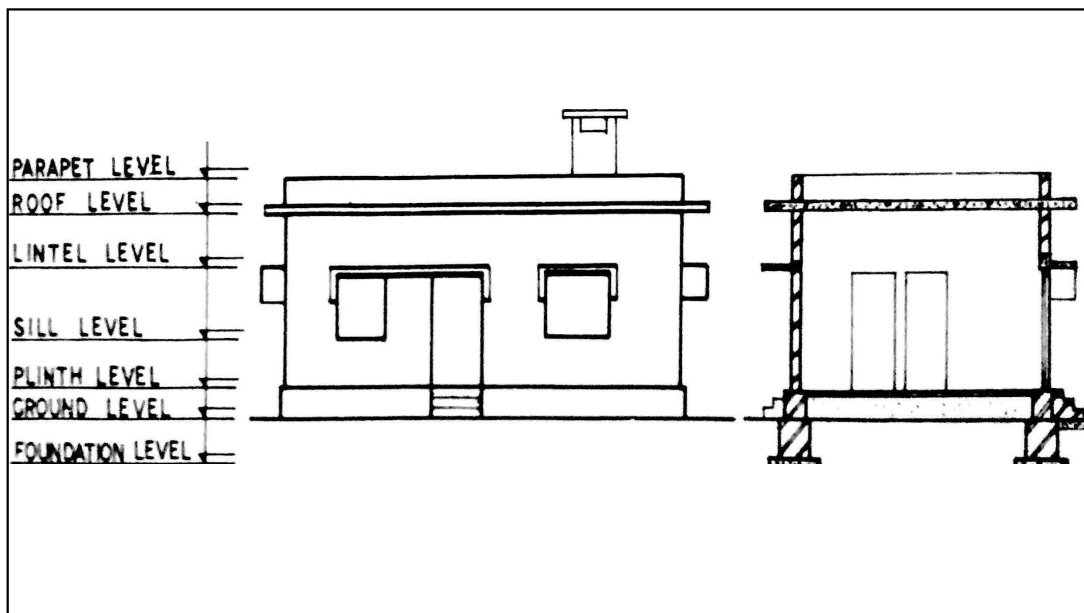
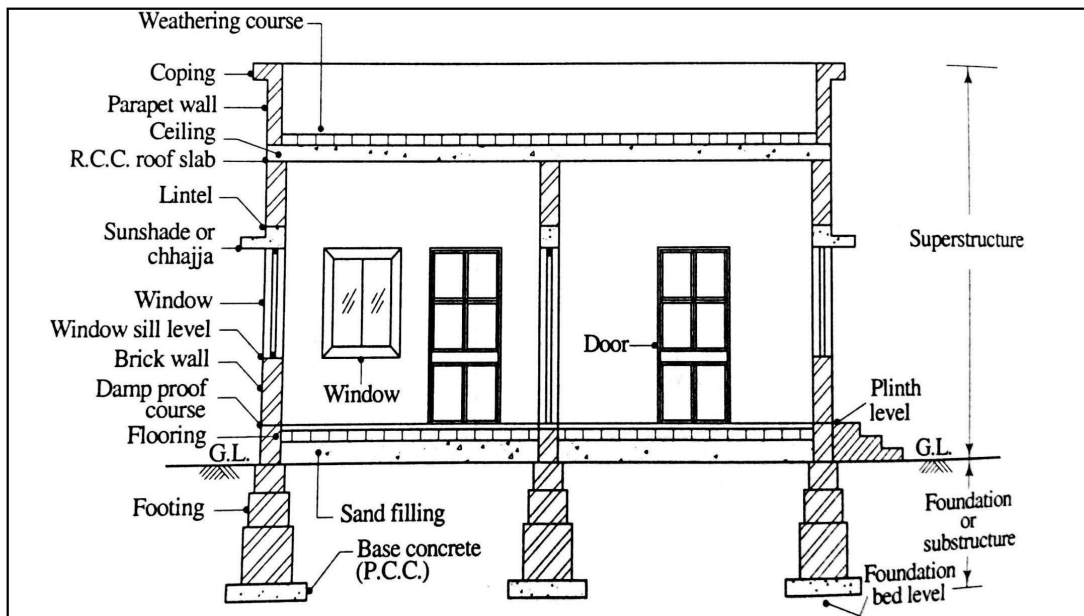


1. **Sub Structure:** The part of building that is constructed below ground level. This part of building safely transfers the load of building to the underlying soil.
2. **Super Structure:** The part of building that is constructed above the ground level.



Components of Building





FOUNDATION

- Lowest part of a building below the ground level and which is in direct contact with the ground.
- Transfers the load on structure to soil over a bearing area.
- It ensure the stability of the building against sliding, overturning and shear failure
- It provides a level and firm surface for the construction of superstructure.
- It prevents unequal or differential settlement of the structure.
- It ensure the stability of the building against undermining due to floodwater or burrowing animals.
- Provides stability to resist force due to wind, earthquake etc.

PLINTH

- Part between ground level and floor level
- The thickness of the plinth wall depends upon the weight of the superstructure and the width of the foundation concrete.
- The minimum height of the plinth is usually kept as not less than 45cm.



Functions of plinth

- To transmit the load of the superstructure to the foundation.
- To protect the building from dampness or moisture.
- To enhance the architectural appearance of the building.

DAMP PROOF COURSE

- Laid at Plinth level
- To prevent moisture rise from foundation to walls
- Cement mortar mixed with crude oil/ water proofing agent/ or damp proof membranes

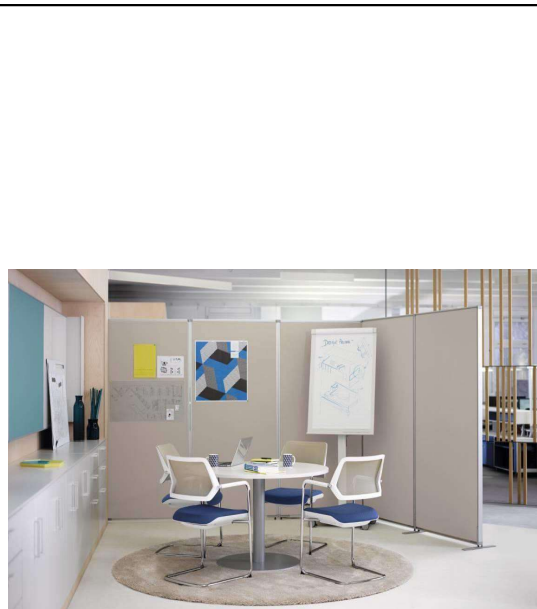


FLOORS

- Provide support for occupants, furniture, and equipments.
- To divide the building into different levels for creating more accommodation within the limited space.
- All the floors, whether basement, ground or upper should be strong enough to support the floor covering and other superimposed loads.
- The floors provide a clean, smooth, impervious, durable and wear-resisting surface.

WALLS

- The main function of walls is to divide the space into different rooms.
- Walls support the loads from the roof/ upper floors to the foundations.
- Piers or pillars are thickened sections of the walls placed at intervals to carry the concentrated loads.
- It should be stable against overturning by lateral forces.
- The external walls should provide sufficient resistance against sun, wind, rain and snow.
- Walls should have sufficient heat and sound insulation.
- Walls should provide sufficient privacy and security against burglary.



DOORS & WINDOWS

- Doors - serve as a connecting link between internal parts and to allow free movement to the outside of the building.
- Windows - proper ventilation and lighting and their number should be determined according to the requirements.
- Strong enough to resist the adverse effects of weather.
- They should not be affected by white ants and the moisture penetration as this will reduce the strength and durability
- They should offer sufficient privacy without inconvenience or trouble and security against theft



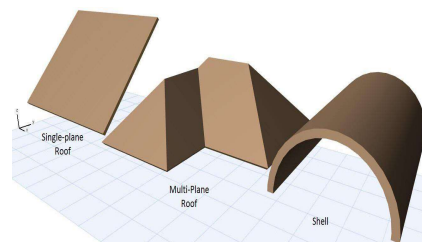
BEAMS, LINTELS & SUNSHADES

- Beam is a horizontal structural member, which carries floor slab or roof.
- Lintel is a beam that supports the masonry work over openings in the walls.
- Sunshade - projection provided outside a building above the doors and windows to prevent direct sunlight and rains to the rooms.
- Window sills are provided to between the bottom of the window frame and the wall below to protect the wall from wear and tear.

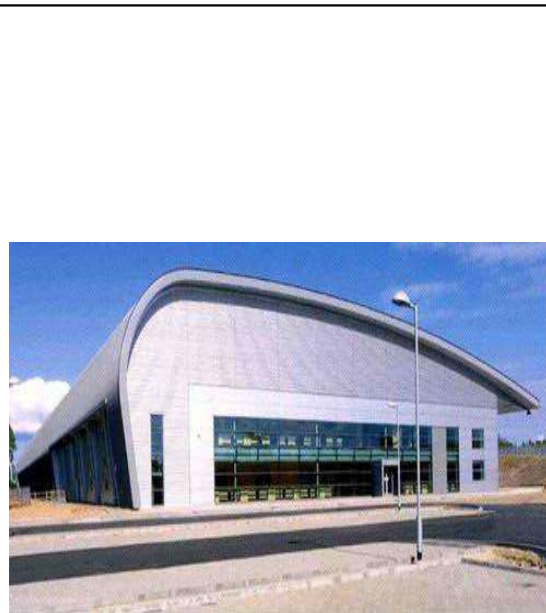


ROOFS

- A roof is the uppermost part of a building whose main functions is to enclose the space and to protect the same from the effects of weather elements.
- Flat or sloped or curved
- Concrete, clay tile, asbestos, iron sheets ,steel truss, timber
- Should be strong against effects of wind, sun , rain
- Drainage

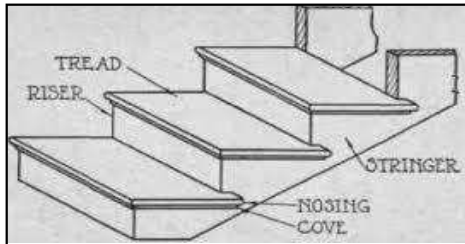


- The roof structure should be strong and stable enough to take up the anticipated loads safely.
- The roof covering should have adequate resistance to resist the effects of weather elements.
- The roof should provide adequate insulation against heat
- The roof should have adequate insulation against sound from external sources.
- Should offer an adequate degree of fire resistance



STEPS & STAIRS

- Steps are provided for access to the building
- A stair is a structure consisting of a number of steps leading from one floor to another
- Location of stairs in all types residential and public buildings should be such as afford the easiest and quickest service possible to the building
- The main function of the stairs is firstly to provide a means of communication between the various floors
- Secondly, it also acts as an escape from the upper floors in the event of fire



PARAPET

- A short masonry wall built on top of the roof of a building is called parapet. It serves as an enclosure above the roof and as an element for good appearance.

WEATHERING COURSE

- It is the layer provided over the roof slab to protect the roof from weathering agencies like sunlight, rain and wind



FINISHES OF WALLS

- Finishes for walls are pointing, plastering, painting, varnishing, polishing etc.
- These finishes protect walls from effects of weather
- It covers the defective materials or poor workmanship to some extent.
- It improves the aesthetic appearance of the building



Plastering

Plaster covering which is applied on the surface of walls and ceilings



Pointing

Finishing of mortar joints



MODULE 1

- Building rules and regulations: Relevance of NBC, KBR & CRZ norms (brief discussion only).
- Building area: Plinth area, built up area, floor area, carpet area and floor area ratio for a building as per KBR.

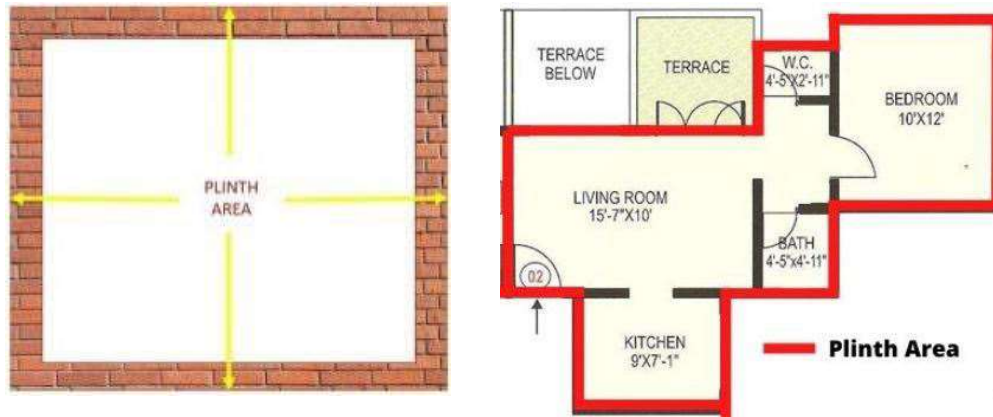
TERMS –BUILDING AREA

- **Plot Area:** Area enclosed within the boundaries of the plot
- **Carpet Area:**
 - ❖ It is the covered area of the usable spaces of rooms at any floor.
 - ❖ Carpet area usually means anything inside the outer walls of an apartment, but excluding the balconies, verandah, toilets, wall thickness or open terrace, electrical and air conditioning plant rooms.
 - ❖ It refers to the area inside your house on which you can actually lay a carpet and physically move around.

TERMS –BUILDING AREA

- **Plinth Area:** Area of the building at the plinth level, does not include the area of open porch uncovered staircase
 - ❖ It is also called as built-up area
 - ❖ It is the entire area occupied by the building including internal and external walls
 - ❖ Plinth area is generally 10-20% more than carpet area.
- **Floor Area:** The usable covered area of the building at any floor level. To get floor area, the area of wall shall be deducted from the plinth area.

TERMS –BUILDING AREA



TERMS –BUILDING AREA

- **Total Floor Area:** Total built up area on all floors excluding parking area
- **Floor Area Ratio:** The ratio of the total floor area on all floors by the area of the plot
 - $\text{FAR} = \text{Total floor area of the floors} / \text{Plot area}$
 - For residential buildings permissible FAR is 3.0

NATIONAL BUILDING CODE

- **The National Building Code (NBC)** is a document that provides guidelines for construction of structures – residential, mercantile, institutional, educational, commercial, assembly, storage spaces or even hazardous buildings.
- National Building Code is prepared to unify the building regulations throughout the country
- NBC is prepared by Bureau of Indian Standards (BIS)

NATIONAL BUILDING CODE

- First version is published in 1970
- This is revised in 1983. Thereafter three major amendments were issued to the 1983 version, two in 1987 and the third in 1997
- Later this edition was again revised to current one in 2005, to which two amendments were issued in 2015.
- Third revision in 2016

RELEVANCE OF NBC

- National instrument providing guidelines for regulating the building construction activities across the country.
- Model code for adoption by all agencies involved in building construction works of PWD, other Govt. Agencies, Private agencies.
- The code contains administrative regulations, development control rules and general requirements, fire safety requirements, stipulations regarding materials, structural design and construction, building services etc.

RELEVANCE OF NBC

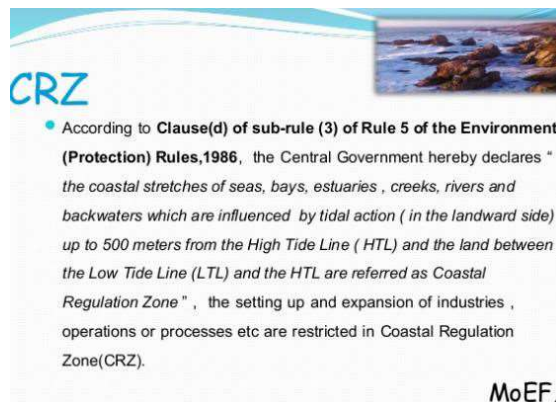
- Building code exists to protect the public's health, safety and welfare
- The building that does not satisfy building code or violation of NBC will lead to penalty, cancellation of sanction or demolition of the building.

COASTAL REGULATION ZONE

- Coastal zone – transition between marine and terrestrial
- Includes coastal beaches, mud flats, wet lands, sea grass ecosystem
- They restrict certain kinds of activities — like large constructions, setting up of new industries, storage or disposal of hazardous material, mining, reclamation and bunding — within a certain distance from the coastline.
- It was implemented on 1991 by Ministry of Environment and Forest (MoEF)

COASTAL REGULATION ZONE

- After the passing of the Environment Protection Act in 1986, CRZ Rules were first framed in 1991.



CLASSIFICATION OF COASTAL REGULATION ZONE

- For regulation of developmental activities, the coastal stretches within 500m of HTL on the landward side are classified into four categories, viz..
- Category I (CRZ-I)
 - Category II (CRZ - II)
 - Category III (CRZ-III)
 - Category IV (CRZ-IV)

CLASSIFICATION OF COASTAL REGULATION ZONE

• CRZ-1:

- ❖ These are ecologically sensitive areas that are essential in maintaining the ecosystem of the coast.
- ❖ such as national parks/marine parks, sanctuaries, reserve forests, wild habitats, mangroves, corals/coral reefs, area close to breeding and spawning grounds of fish and other marine life, areas of outstanding natural beauty, historical and heritage areas, areas rich in genetic biodiversity
- ❖ They lie between low and high tide line.
- ❖ Exploration of natural gas and extraction of salt are permitted

CLASSIFICATION OF COASTAL REGULATION ZONE

• CRZ-2:

- ❖ These areas are urban areas located in the coastal areas

• The areas that have already been developed up to or close to the shoreline. For this purpose, "**developed area**" is referred to as that area within the municipal limits or in other legally designated urban areas which is already substantially built up and which has been provided with drainage and approach roads and other infrastructural facilities, such as, water supply and sewerage mains.

- ❖ Under Coastal Zone Regulation (CRZ) Notification 2018, the floor space index norms has been de-frozeed.

CLASSIFICATION OF COASTAL REGULATION ZONE

• CRZ-3:

• Areas that are relatively undisturbed and those which do not belong to either Category-I or II. These will include coastal zone in the rural areas (developed and undeveloped) and also areas within municipal limits or in other legally designated urban areas which are not substantially built up.

- ❖ Only certain activities related to agriculture even some public facilities are allowed in this zone

CLASSIFICATION OF COASTAL REGULATION ZONE

- **CRZ-4:**

- ❖ This lies in the aquatic area up to territorial limits.
- ❖ Fishing and allied activities are permitted in this zone.
- ❖ No Solid waste should be let off in this zone.
- ❖ This zone has been changed from 1991 notification, which covered coastal stretches in islands of [Andaman & Nicobar](#) and [Lakshadweep](#).

Kerala Building Rules(KMBR/KPBR)

- Building construction in Kerala is regulated by,
 - Kerala Municipality Building Rules,1999-Apply to all Municipalities
 - Kerala Panchayath Building Rules,2011-Apply to all Village Panchayath
- Prior to that, construction of buildings was being regulated by Kerala Building Rules, 1984(KBR).

- According to KMBR/KPBR, any person intends to construct/reconstruct a building should submit an application along with site plan ,ground plan, elevation, sections of the building and specifications of the work to the Secretary of the Municipal Corporation / Municipality / Panchayat.
- The applicant has to submit documents to prove ownership of the land concerned and payment of application fee along with a copy of the certificate of registration of the architect / building designer.
- The Secretary inspects site, verifies whether the building plan, elevation and sections of the buildings and specifications of the work conform to the site and site plan and is in accordance with KMBR/KPBR.

- Then Secretary approves the plan and issues permit to execute the work on remittance of the permit fee at the prescribed rates.
- He / she also has the power to refuse approval or to demand modifications to the plan which should be communicated in writing.
- Constructions of Central and State Government such as Railways, National Highways, Waterways, Aerodromes, etc., are exempted from KMBR / KPBR.
- Similarly, permits are not necessary for minor works such as providing and removing windows, doors and ventilators for partition, painting, petty repairs etc.
- The Rules also prescribe specific and separate norms for parking spaces, open area, fire escape, ventilators, sanitation facilities, front and rear yards, etc., for different type of buildings based on occupancy.